

Appendix to Cross-Sectional Factor Momentum: Evidence from Multiple Formation Periods

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Abstract

This appendix supplements tables to Cross-Sectional Factor Momentum: Evidence from Multiple Formation Periods.

Keywords: Cross-sectional factor momentum; Formation periods; Publication effect

JEL Classification: G12; G14

Table A1. Alphas of Portfolios Sorted by r_1 and $r_{7,12}$

This table extends Table 3 with the annualized alphas of the factor momentum portfolios. Each portfolio is regressed on the market factor (CAPM), the Fama–French three factors (FF-3), and the Fama–French five factors (FF-5). CAPM and FF-3 regressions span July 1926 to December 2023, and FF-5 regressions span July 1963 to December 2023. In parentheses are Newey–West t-statistics with 12 lags, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Single-sort by r_1

	Loser	Mid	Winner	WML
α by CAPM	-0.08 (-0.09)	3.46*** (10.48)	6.12*** (7.91)	6.20*** (4.18)
by FF-3	0.44 (0.51)	3.63*** (10.34)	5.88*** (8.33)	5.44*** (3.85)
by FF-5	-1.89** (-2.46)	1.37*** (5.75)	4.25*** (5.13)	6.14*** (4.04)

Panel B. Single-sort by $r_{7,12}$

	Loser	Mid	Winner	WML
α by CAPM	1.63** (2.16)	3.11*** (7.30)	4.78*** (5.92)	3.15** (2.23)
by FF-3	1.49** (2.35)	3.32*** (8.11)	5.17*** (6.44)	3.67*** (2.87)
by FF-5	0.05 (0.07)	1.23*** (3.75)	2.46*** (3.36)	2.41* (1.76)

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Table A1. *Continued.* Alphas of Portfolios Sorted by r_1 and $r_{7,12}$ Panel C. Independent double-sort by r_1 and $r_{7,12}$

		$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
α by CAPM	r_1 Loser	-0.26 (-0.24)	0.01 (0.01)	1.13 (1.15)	0.95 (0.79)
	r_1 Mid	2.30*** (3.22)	3.14*** (6.32)	4.77*** (6.41)	2.41* (1.93)
	r_1 Winner	4.36*** (4.09)	5.89*** (7.82)	7.73*** (8.47)	3.03** (2.52)
	r_1 WML	4.69*** (2.58)	5.90*** (4.07)	6.39*** (4.85)	
by FF-3	r_1 Loser	-0.15 (-0.14)	0.39 (0.44)	1.78* (1.81)	1.40 (1.25)
	r_1 Mid	2.34*** (3.62)	3.34*** (6.63)	5.10*** (7.11)	2.70** (2.37)
	r_1 Winner	3.83*** (4.16)	5.89*** (8.16)	7.78*** (8.20)	3.48*** (3.01)
	r_1 WML	4.19** (2.37)	5.53*** (4.16)	6.31*** (4.76)	
by FF-5	r_1 Loser	-2.15** (-2.20)	-1.90*** (-2.83)	-0.81 (-0.96)	1.00 (0.82)
	r_1 Mid	1.08* (1.89)	0.96** (2.58)	1.62*** (2.78)	0.54 (0.52)
	r_1 Winner	3.40*** (3.71)	4.25*** (5.52)	5.58*** (5.38)	1.79 (1.38)
	r_1 WML	5.80*** (3.94)	6.15*** (5.01)	6.53*** (5.16)	

Table A1. *Continued.* Alphas of Portfolios Sorted by r_1 and $r_{7,12}$ Panel D. Dependent double-sort by r_1 first and then $r_{7,12}$

		$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
α by CAPM	r_1 Loser	-0.91 (-0.81)	-0.13 (-0.13)	0.81 (0.87)	1.73 (1.50)
	r_1 Mid	2.44*** (3.42)	3.04*** (7.15)	4.99*** (6.37)	2.56* (1.96)
	r_1 Winner	3.95*** (3.90)	6.41*** (8.14)	7.69*** (7.84)	3.74*** (3.29)
	r_1 WML	4.86*** (2.68)	6.54*** (4.41)	6.87*** (4.68)	
by FF-3	r_1 Loser	-0.57 (-0.49)	0.43 (0.46)	1.43 (1.63)	2.00* (1.86)
	r_1 Mid	2.58*** (4.17)	3.14*** (7.02)	5.28*** (6.88)	2.70** (2.33)
	r_1 Winner	3.41*** (3.70)	6.33*** (8.67)	7.55*** (8.00)	4.14*** (3.72)
	r_1 WML	3.98** (2.18)	5.90*** (4.24)	6.13*** (4.52)	
by FF-5	r_1 Loser	-2.47** (-2.25)	-2.13*** (-2.86)	-1.08 (-1.25)	1.39 (1.26)
	r_1 Mid	1.16** (2.16)	1.21*** (3.04)	1.75*** (3.14)	0.59 (0.59)
	r_1 Winner	3.19*** (3.37)	4.05*** (5.27)	5.52*** (4.81)	2.33** (2.09)
	r_1 WML	5.66*** (3.24)	6.17*** (4.58)	6.61*** (4.13)	

Table A1. *Continued.* Alphas of Portfolios Sorted by r_1 and $r_{7,12}$ Panel E. Dependent double-sort by $r_{7,12}$ first and then r_1

		$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
α by CAPM	r_1 Loser	-0.54 (-0.48)	0.50 (0.55)	1.91* (1.79)	2.45 (1.60)
	r_1 Mid	1.65* (1.88)	3.14*** (5.97)	4.77*** (5.63)	3.12** (2.04)
	r_1 Winner	3.69*** (3.58)	5.64*** (7.59)	7.65*** (7.91)	3.95*** (2.84)
	r_1 WML	4.23*** (2.64)	5.14*** (3.70)	5.73*** (4.77)	
by FF-3	r_1 Loser	-0.48 (-0.44)	0.86 (1.04)	2.36** (2.43)	2.84** (2.13)
	r_1 Mid	1.69** (2.22)	3.41*** (6.50)	5.32*** (5.84)	3.63** (2.53)
	r_1 Winner	3.20*** (3.69)	5.64*** (7.90)	7.81*** (8.00)	4.60*** (3.50)
	r_1 WML	3.69** (2.38)	4.78*** (3.77)	5.45*** (4.71)	
by FF-5	r_1 Loser	-3.02*** (-2.95)	-1.72*** (-2.60)	-0.52 (-0.62)	2.50* (1.85)
	r_1 Mid	0.53 (0.70)	1.13*** (3.21)	2.15*** (2.80)	1.62 (1.12)
	r_1 Winner	2.63*** (2.93)	4.27*** (5.74)	5.73*** (5.18)	3.10*** (2.06)
	r_1 WML	5.65*** (4.20)	5.99*** (5.33)	6.24*** (4.94)	

Table A2. Spanning Alphas of Portfolios Sorted by r_1 and $r_{7,12}$

This table reports the annualized spanning alphas of the recent- (r_1 WML) and intermediate-past ($r_{7,12}$ WML) factor momentum strategies, with each strategy regressed on the other as follows:

$$\begin{aligned} \text{WML}_t^1 &= \alpha^1 + \beta^1 \text{WML}_t^{7,12} + \varepsilon_t^1 \\ \text{WML}_t^{7,12} &= \alpha^{7,12} + \beta^{7,12} \text{WML}_t^1 + \varepsilon_t^{7,12}. \end{aligned}$$

Each regression spans July 1927 to December 2023. In parentheses are Newey–West t-statistics with 12 lags, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Parameter	r_1 WML	$r_{7,12}$ WML
α	6.34*** (3.09)	3.08** (2.47)
β	-0.02 (-0.13)	-0.02 (-0.13)
Adjusted r-squared	-0.0004	-0.0004
Number of observations	1,158	1,158

Table A3. Performance of Dependently Double-Sorted Portfolios by r_1 and $r_{7,12}$

This table reports the annualized average monthly returns of equal-weighted tercile factor momentum portfolios sorted by the past 1- (r_1) and (7-12)-month ($r_{7,12}$) returns of factors, as a continuation of Table 3 in the main document. Panel A (B) dependently sorts 3-by-3 portfolios by r_1 ($r_{7,12}$) first and then $r_{7,12}$ (r_1). The factors span January 1926 to December 2023; accordingly, portfolios sorted by r_1 ($r_{7,12}$) begin in February 1926 (January 1927). t-statistics are in parentheses, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Dependent double-sort by r_1 first and then $r_{7,12}$

Sorted Vertically First by	$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
r_1 Loser	-2.15** (-1.98)	-1.83* (-1.80)	-0.46 (0.43)	1.69 (1.52)
r_1 Mid	0.95 (1.25)	1.81*** (3.36)	3.76*** (4.54)	2.81** (2.30)
r_1 Winner	3.03*** (2.73)	4.67*** (5.02)	6.41*** (5.62)	3.38*** (2.74)
r_1 WML	5.18*** (2.85)	6.50*** (4.24)	6.87*** (4.21)	

Panel B. Dependent double-sort by $r_{7,12}$ first and then r_1

Sorted Horizontally First by	$r_{7,12}$ Loser	$r_{7,12}$ Mid	$r_{7,12}$ Winner	$r_{7,12}$ WML
r_1 Loser	-1.60 (-1.52)	-0.80 (-0.95)	0.78 (0.70)	2.38 (1.53)
r_1 Mid	0.10 (0.11)	1.62** (2.58)	3.15*** (3.17)	3.05** (2.06)
r_1 Winner	2.57** (2.38)	4.11*** (5.20)	6.14*** (5.36)	3.56** (2.34)
r_1 WML	4.17*** (2.85)	4.91*** (3.96)	5.36*** (4.49)	

Table A4. Fama–French Factor Loadings of Stock Momentum, Recent-Past Factor Momentum, and Intermediate-Past Factor Momentum

This table reports the Fama–French factor loadings of the stock momentum ($r_{2,12}$ WML), recent-past factor momentum (r_1 WML), and intermediate-past factor momentum ($r_{7,12}$ WML) strategies, with each strategy regressed as follows:

$$\text{WML}_t = \beta' \mathbf{f}_t + \varepsilon_t,$$

where $\mathbf{f}_t$ is either the Fama–French three or five factors. Consistent with the factor momentum strategies, the stock momentum strategy buys (sells) the value-weighted top (bottom) 30% portfolio sorted by previous (2-12)-month returns. The Fama–French three factors span July 1926–November 2024, and the five factors do July 1963–November 2024. t-statistics are in parentheses, and ***, **, and * mean significance at the 1%, 5%, and 10% levels, respectively.

Variable	Stock	Recent	Intermediate	Stock	Recent	Intermediate
Intercept	10.78*** (6.75)	5.44*** (3.47)	3.67*** (2.61)	7.00*** (3.47)	6.14*** (3.59)	2.41* (1.67)
Mkt-RF	-0.24*** (-9.04)	-0.04 (-1.44)	-0.03 (-1.27)	-0.17*** (-4.20)	-0.11*** (-3.25)	-0.03 (-0.90)
SMB	-0.08* (-1.77)	0.06 (1.34)	0.23*** (6.03)	-0.03 (-0.50)	0.08 (1.52)	0.06 (1.32)
HML	0.47*** (12.48)	0.24*** (6.54)	-0.24*** (-7.30)	-0.54*** (-7.18)	0.01 (0.19)	-0.33*** (-6.13)
RMW				0.16** (1.99)	0.11 (1.59)	-0.05 (-0.88)
CMA				0.36*** (3.13)	0.21** (2.14)	0.21** (2.53)
Adjusted r-squared	0.2214	0.0356	0.0675	0.0985	0.0395	0.0553
Number of observations	1,175	1,158	1,158	737	720	720

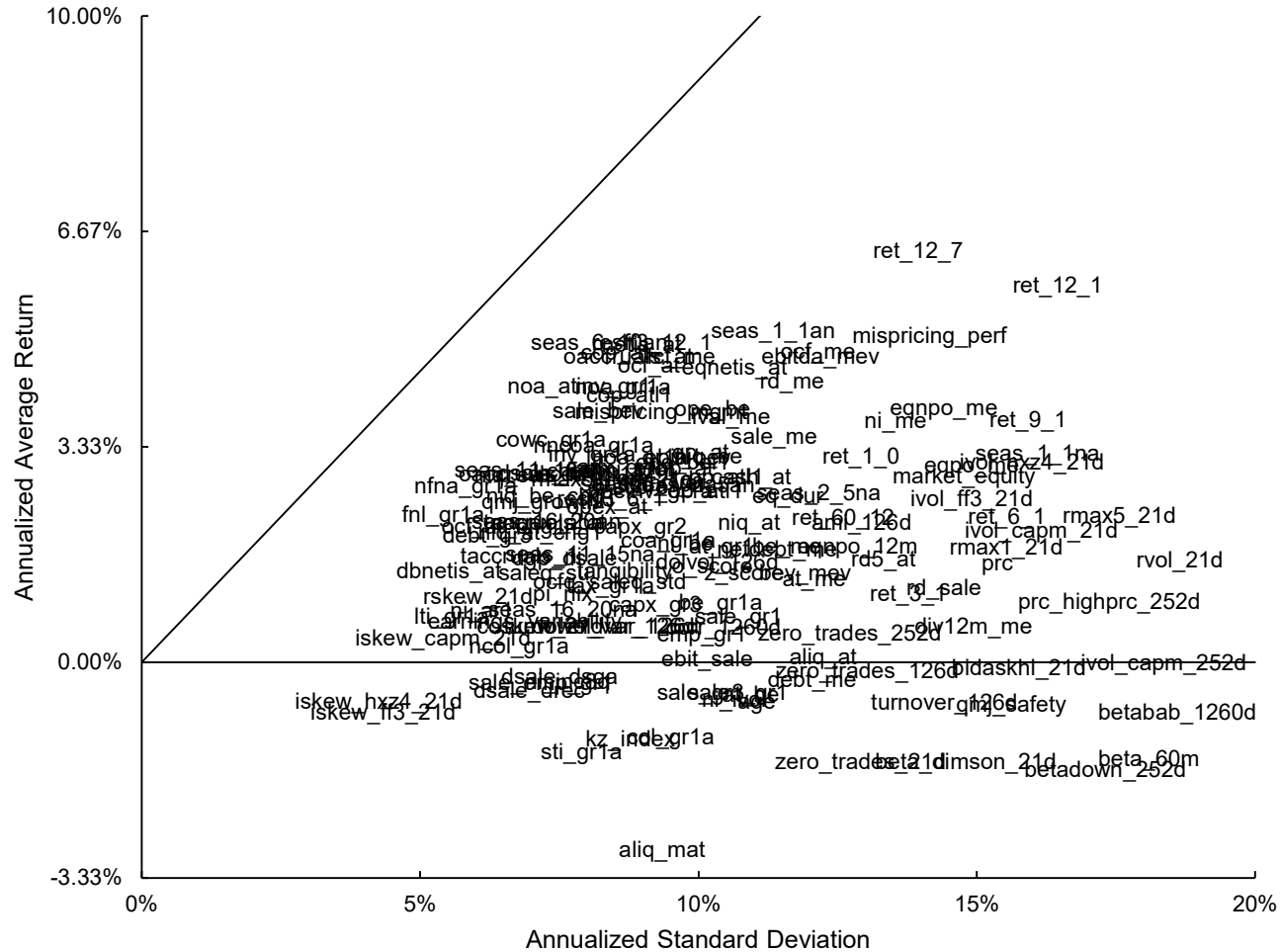


Figure A1. Capital Market Line and Jensen–Kelly–Pedersen Factors

This figure shows the capital market line by the Fama–French market factor from July 1926 to December 2023 and the annualized standard deviations (x-axis) and average returns (y-axis) of 153 U.S. value-weighted monthly factors, spanning January 1926 to December 2023. Each factor represents a long-short strategy that buys (sells) the top (bottom) tercile portfolio sorted by a given characteristic.

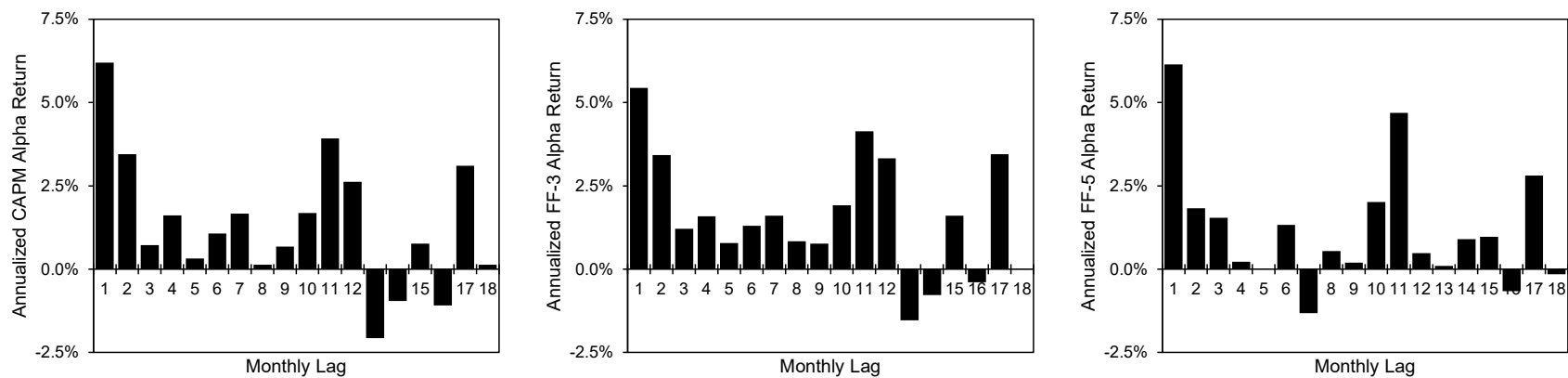


Figure A2. Single-Month Strategy Alphas

This figure shows the annualized alpha returns for strategies based on single-month performance. Each strategy is regressed on the market factor (CAPM, left), the Fama–French three factors (FF-3, center), and the Fama–French five factors (FF-5, right). Winners and losers are the top- and bottom-tercile portfolios, respectively. Jensen–Kelley–Pedersen factors span January 1926 to December 2023. CAPM and FF-3 regressions span July 1926 to December 2023, and FF-5 regressions span July 1963 to December 2023.